

REGULAR PAPER

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In-vehicle augmented reality system to provide driving safety information

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Abstract Improving traffic safety is one of the important goals of intelligent transportation systems. Traffic signs play a very vital role in safe driving and in avoiding accidents by informing the driver about the speed limits or possible dangers such as icy roads, imminent road works or pedestrian crossings. In-vehicle contextual Augmented reality (AR) has the potential to provide novel visual feedbacks to drivers for an enhanced driving experience. In this paper, we propose a new AR traffic sign recognition system (AR-TSR) to improve driving safety and enhance the driver's experience based on the Haar cascade and the bag-of-visual-words approach, using spatial information to improve accuracy and an overview of studies related to the driver's perception and the effectiveness of the AR in improving driving safety. In the first step, the region of interest (ROI) is extracted using a scanning window with a Haar cascade detector and an AdaBoost classifier to reduce the computational region in the hypothesis-generation step. Second, we proposed a new computationally efficient method to model global spatial distribution of visual words by taking into consideration the spatial relationships of its visual words. Finally, a multiclass sign classifier takes the positive ROIs and assigns a 3D traffic sign for each one using a linear SVM. Experimental results show that the suggested method could reach comparable performance of the state-of-the-art approaches with less computational complexity and shorter training time, and the AR-TSR more strongly impacts the allocation of visual attention during the decision-making phase.

Keywords Augmented reality · Traffic sign recognition · SVM · Haar cascade · Bag-of-visual-words · Driving safety · Intelligent transportation systems

1 Introduction

In recent years, the use of the automobile as the primary mode of transportation has increased significantly and driving has become an important activity of daily life. As such, road traffic generates serious problems in terms of congestion, safety, and environmental impact. Today, due to the rapidly growing number of vehicles, road accidents constitute the eighth leading cause of death. Most traffic accidents have been caused by driver's oversight of important objects such as pedestrians, traffic signs, traffic signals, and so on. Research shows that human errors including driver inattention or cognitive overload lead to misjudgments and delays in environment recognition and constitute a major factor of road accidents. Even though some

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developments in passive safety technologies, such as seatbelts, airbags, and crumple zones have partially reduced damages and improved safety during accidents, further progress in these technologies is limited due to their inherited limitations (Shahid et al. 2013). With the development of information technology, there are a lot of advanced electronic devices used for driving safety assistance (Wang et al. 2011).

Automotive active safety systems have become increasingly common in road vehicles since they provide an opportunity to significantly reduce traffic fatalities by active vehicle control. Traffic signs play a very vital role in safe driving and in avoiding accidents by informing the driver of speed limits or possible hazards. In the real world, knowing the content of the sign is much more important than simply knowing its existence. In such systems, a video camera is installed in the interior of the vehicle and the environment is scanned to provide useful information for the driver. In fact, an increasing effort has been dedicated to developing driver-assistance systems based on computer vision in the past years, aiming to reduce the number and fatality of traffic accidents, focusing on several aspects such as pedestrian detection (Geronimo et al. 2010; Guo et al. 2012), lane detection/lane departure warning systems (McCall and Trivedi 2006; Wang et al. 2010), or traffic sign detection (TSD)/traffic sign recognition (TSR) (Park and Kim 2013; Mogelmose et al. 2012).

Extensive studies in human-machine interactivity are necessary to present the TSR information in a careful way to inform the driver without causing distraction or confusion. The visibility of traffic signs is crucial for the drivers' safety. In a real environment, it becomes a difficult task to recognize the traffic signs timely and accurately because the visibility of traffic signs may be decreased greatly by some unfavorable factors. For example, very serious accidents happen when drivers do not notice a stop sign. An automatic TSR system is helpful for assisting drivers and is essential for autonomous cars (Azad et al. 2014).

Fatigue, divergence of attention, and occlusion of signs due to road obstruction and natural scenes may lead to miss some important traffic signs, which can result in severe accidents. Guiding the driver's attention to an imminent danger, somewhere around the car, is a potential application. In cars, the augmented reality (AR) is becoming an interesting means to enhance the active safety in the driving task. To acquire traffic information and keep the driver's eye on roads at the same time, a drastic increase in the developments of the AR applications on a head-up display (HUD) has been noticed. After recognizing the traffic signs, a driver may be notified of the recognized traffic signs in a manner of audio or visual information.

In-vehicle contextual AR has the potential to provide novel visual feedback of other automatic functionalities to drivers for an enhanced driving experience such as traffic signs recognition, lane deviation warnings, safety distance indication, and forward collision warnings. The AR-HUD technologies aim to optimize the visual attention of the driver by increasing the salience of high-value elements and to enhance the intelligent transportation systems by superimposing surrounding traffic information on the users' view and by keeping the drivers' view on roads. However, due to the existence of a complex environment such as weather conditions, illuminations and geometric distortions, the AR-HUD TSR systems have always been considered as a challenging task. Although traffic signs are designed to be clearly visible, they can be missed due to driver distraction or sign masking.

Road scenes are also generally very cluttered and contain many strong geometric shapes that could easily be misclassified as road signs. Achieving robust TSR with short processing times is a very challenging endeavor (Yin et al. 2015). Such a vision-based safety system has remained elusive in cars because computers typically face a tradeoff between analyzing video images quickly and drawing the right conclusions. In conventional techniques, color, shape information, or geometric features of traffic signs are utilized for performing detection and recognition with regard to the traffic signs in general. Road sign recognition systems usually have developed into two specific phases. First, in each frame, the detection stage identifies also the categories of signs based on shapes (circular, rectangular, triangular, etc.). The second task is to classify the detected signs and send the processing results (i.e., the types of signs and their locations) to the display and control units of an Advanced Drive Assistance System (ADAS).

There are several challenges involved in developing a complete TSR system that includes TSD and classification. These include occlusion of signs due to different background, weather condition, viewpoints and sign deformations. To achieve fast and robust TSD, designing a computing efficient and highly discriminative feature is essential. Besides, classification of traffic signs is a complicated matter since sign types are similar. Recently, the bag-of-visual-words (BoVW) has been frequently used in the classification of image data. There is a significant amount of work which present interesting advances for creating better dictionaries (Nguyen et al. 2013; Virupakshappa et al. 2015).

In the traditional BoVW model, spatial information between keypoints is ignored during visual word construction when using simple clustering algorithms such as k-mean. However, one major limitation of the

standard BoVW model is that it ignores spatial information of visual words in image presentation and comparison. Researchers have demonstrated that the object recognition performance can be improved by including spatial information, which is important for similarity measurement between images (Zhu et al. 2016; Li et al. 2016, 2015). Therefore, combining the frequency of occurrence and spatial information of visual words is a promising direction for improving classification accuracy.

The recognition of road traffic signs correctly at the right time for that particular place is very important to ensure a safe journey. To achieve fast and robust TSR, we introduce a novel way to incorporate both distance and angle information in the BoVW representation. Considering the processing time and classification accuracy as a whole, we have developed a novel technique to incorporate spatial information of visual words to improve accuracy while maintaining short retrieval time.

Another important issue is the study of conventional traffic signs, in terms of rules for placement and visibility, types of traffic signs, and the migration of these to an in-vehicle display. In this paper, a novel method is proposed for the TSR based on the Haar cascade and the BoVW approach, using spatial information to improve accuracy. We also provide an overview of studies related to the driver's perception and the effectiveness of the AR in improving driving safety. The AR-TSR is expected to make driving even more comfortable while helping drivers receive important information regarding the signs, even before their eyes can actually see the sign.

We present two key contributions. First, to improve driving safety and enhance the driver's experience, we propose a new AR-TSR system that displays visual cues on the drivers' view while keeping drivers view on roads. Second, a novel approach for visual word construction is presented, which takes the spatial information of keypoints into account to enhance the quality of visual words generated from extracted keypoints. We demonstrate the complementarities of the additional relative spatial information provided by our approach to improve accuracy while maintaining short retrieval time.

The rest of our work is organized as follows. First, an overview of related works will be presented in Sect. 2. Then, our AR-TSR method is proposed in Sect. 3. After that, Sect. 4 deals with the experimental results, discussed afterwards to evaluate them. Finally, conclusion is provided in Sect. 5.

2 Related work

Vehicular safety has been actively explored in the recent years. In fact, even before the appearance of motorized vehicles a lot of devices had been developed and placed in vehicles (Silvéria 2014). The design of TSR has been a challenging problem for many years, hence becoming an important and active research topic in the area of intelligent transport systems. Traffic sign localization and classification form a base for advanced methods used for accurate TSR and autonomous vehicle driving so that traffic accidents can be prevented and safety of traffic participants can be increased.

In this regard, such technologies and products have been developed around the world, and driving safety information is now being offered to the driver through various modalities. Thus, driving workloads including visual, auditory, psychomotor, and cognitive arise when the driver recognizes information transferred through touch sensors, in-vehicle terminals, and so on (Ho and Spence 2005). Accordingly, the ability to recognize signs and their underlying information is highly desirable.

Furthermore, several approaches and techniques for road sign detection and recognition have been published (Mogelmose et al. 2012; Chuan et al. 2011). The most common approach, quite sensibly, consists of two main stages: detection and recognition. The considered baseline algorithms represent some of the most popular detection approaches such as the Viola–Jones detector, based on Haar features (Jeon et al. 2014), and the linear classifier relying on the histogram of oriented gradients (HOG) descriptors. Some recent methods, such as those by Greenhalgh and Mirmehdi (2012) and Kun et al. (2014) have used the HOG features for road sign feature extraction, using complementary features to reduce the computation complexity of TSD, and then using the SVM to implement the traffic sign classification.

Moreover, the convolutional neural network (CNN) has been adopted in object recognition for its high accuracy. In the study by Stallkamp et al. (2012), an approach based on the combination of color transformation and multi-layer CNNs were proposed using a combination of supervised and unsupervised learning. This model could learn multi-stages of invariant features of image, with each layer containing a filter bank layer, a non-linear transform layer, and a spatial feature pooling layer. Shape and color information from the detection stage was used to select the appropriate neural network. This method is still not accurate and fast enough for the ADAS application.

In Sermanet and LeCun (2011), they applied Convolutional Networks (ConvNets) to the task of traffic sign classification. The ConvNets were biologically inspired multi-stage architectures that automatically learned hierarchies of invariant features. The CNNs consist of a multi-stage processing of an input image to extract hierarchical and high-level feature representations. In Cireşan et al. (2012) a real-time system for traffic signs was put forward, which used a sliding window method combining various DNNs trained on differently preprocessed data into a Multi-Column DNN (MCDNN). The latter further boosted recognition performance, where the method transformed the original image into the gray-scale image using support vector machines. Then it used CNNs with fixed and learnable layers for detection and recognition.

Various approaches have been proposed to solve the problem of TSD. In Houben et al. (2013), a shape-based detector and classifier are used. First, the detector roughly finds out all candidate region of interests (ROIs) using HOG features and second, the candidate ROIs are resized to 40 x 40 windows and further verified. Ultimately, LDA and IK-SVM are used for classification. In Mathias et al. (2013), the authors proposed a method that exploits specific shapes and colors of traffic signs. The proposed method works by computing features similar to Haar-like features in the integral image of several channels of the input image. The integral channel feature classifier is a family of boosted classifiers based on discrete Adaboost. In Liang et al. (2013), a combination of HOG features and color histogram features is used for TSR. First, a set of ROIs is extracted using HOG and color histograms. Second, a recognition model is applied to each ROI using the RBF kernel SVM to classify the candidate ROI.

Another TSD framework was presented by Zaklouta et al. (2011) using different size of HOG descriptors and the distance transforms. They combined the tree classifiers with the HOG descriptors, where the K-d tree achieves a classification result of 92.9% with HOG descriptors and 67% with the distance transforms. In Gonzalez-Reyna et al. (2013) a fast detector was used for a round speed sign based on the oriented gradient maps and the Karhunen–Loeve transform to improve the classification performance. Global sign shapes were first detected with various heuristics and color thresholding, then the detected windows were classified using a different multi-layer neural network for each type of outer shape (Sermanet and LeCun 2011).

The above-described approach ignores structural information of features, which is important for similarity measurement between images. It is, therefore, necessary to classify the characteristics of the given information and find a way to represent the information according to these characteristics. Several methods were recently proposed to incorporate spatial information to improve the BoVW model such as the spatial pyramid matching method (Li et al. 2016), spatio-temporal interest point (Dawn and Shaikh 2015), and the distance between joint histograms to measure the similarity between a target and its candidate patches (Li et al. 2015). Other recent works have focused on the local stability of traffic sign regions (Li et al. 2015). Further, methods such as those by Haloi (2015) proposed a novel classification techniques of TSR based on probabilistic latent semantic. The algorithm consists of two parts: (1) classifying the shape of the traffic signs and (2) classifying its actual class. To investigate the effect of coding methods and codebook size of local spatiotemporal features, (Danelakis et al. 2016) have proposed a coding method to alleviate the negative effect of quantization error by assigning the local spatiotemporal features to a few nearest visual words.

In addition, research needs to be conducted on both the amount of information that can minimize a driver's cognitive load and the types of representation that can minimize the driver's visual load. The use of the AR in a vehicular context is then introduced, identifying various projects that have already explored this technology as a way to increase the road safety and the overall driving experience. Traffic signs or road signs are an important part of the road environment as they provide visual messages not only for drivers but also for all road users. Several detection algorithms are based on edge detection, making them more robust to changes in illumination.

Although some research has been performed to evaluate the methods for directing the driver attention using AR cues (Kim and Dey 2009; Tonniss et al. 2005), the primary goal of this study was to determine the costs and benefits of dynamic conformal AR cues to alert experienced drivers to potential roadway hazards. In AR applications, the model-based recognition and 3D pose estimation are often used for superposing computer-generated images over views of the real world in real time. For 3D object visualization, orientation and location values must be calculated. Also, calibrated cameras are needed for AR scenarios, which provide the combination of real-world information and virtual contents by superimposing graphical objects to the real world (Azuma 1997).

The international automotive supplier, Continental, is developing an AR-HUD to facilitate a new form of dialog between the vehicle and the driver. The reality of the traffic conditions, as seen by the driver, is

augmented in this display system, and the graphical images placed precisely into the exterior view by means of an optical projection embed information directly into the traffic situation. This makes the AR-HUD one of the most innovative enhancements of the human-machine interface (HMI) in vehicles (Richter 2014).

Another interesting method was raised by Livingston et al. (2003) who described a study that examined multiple methods for a visually depicting occlusion in an outdoor AR, varying opacity, stroke, and fill settings. The study found that users have difficulty discerning more than a few levels of occluded objects, even though the AR graphic opacity has appeared to be promising as an effective layering and ordering cue. Nevertheless, most research groups have conducted user studies with prototypes in driving simulators due to difficulties in implementing hardware (Doshi et al. 2009). It is shown that an AR graphic conveying a speeding alert has reduced the driver distraction and reaction time to the alerts.

Limited research has been conducted on the effectiveness of AR cuing for TSD. We have already seen some limited, but successful, automotive AR systems in the market today (Schall et al. 2012). A handful of manufacturers offer a video-based AR auxiliary display to aid drivers in maneuvering the vehicle and identifying potential hazards while backing up. As the automotive industry moves toward the car of the future—aiming to make driving safer, more pleasant, and more convenient—we have already seen some successful video-based AR auxiliary displays. However, neither the application opportunities of an optical see-through AR, as presented on a driver's windshield, are fully tapped, nor the visual perceptual and attention challenges are fully understood.

3 Proposed method

In this section, we present an overview of the proposed TSR method via a marker-less AR. The distinctive features of traffic signs should be considered since traffic signs are regulated in specific colors and shapes. It is, therefore, convenient to use those features to decide on the candidate sign. Another important issue was the study of conventional traffic signs, in terms of rules for placement and visibility, types of traffic signs and the migration of these to an in-vehicle display. To address this problem, we propose AR-TSR systems that display visual cues on the drivers' view of the road with the ultimate goal of increasing the driver awareness and safety.

The vision algorithms for driver-assistance systems usually need to fulfill strong real-time constraints. Hence, we draw a particular focus on real-time capability of the algorithms evaluated here. Our detector is inspired by a detector presented by Viola and Jones (2004). In the first step, the ROI is extracted using a scanning window with a Haar cascade detector and an AdaBoost classifier to reduce the computational region in the hypothesis-generation step. Next in the verification phase, to confirm whether each ROI is a traffic sign or not, a second stage is needed to eliminate some false positives. In this stage, with the feature extraction of traffic signs based on the speeded up robust features (SURF), the codebook is generated by this feature clustering and the images are described by histograms using the BoVW for verification. To ensure rotation invariance, we proposed a new computationally efficient method to model global spatial distribution of visual words and improved the standard BoVW representation, by taking into consideration the spatial relationships of its visual words. Finally, a multiclass sign classifier takes the positive ROIs and assigns a 3D traffic sign for each one using a linear SVM.

Figure 1 shows the overall procedure of the marker-less AR system, which is split up into two distinct stages. In the above two stages, we assume that the intrinsic and distortion parameters of the camera are known and do not change, these two stages are detailed in Abdi et al. (2015).

3.1 Generation candidate detection bounding boxes

The initial detection phase of a TSR system has much computational costs because ROIs in a large range of scales have to be searched in the complete image. To reduce the search space, the adopted solution is to combine a cascade with fewer stages with other methods that eliminate the false positives. During the detection phase, the system scans each window of the input image and extracts the Haar-like features of that particular window, which is then used to compare to the cascade classifier. Finally, only a few of these sub-windows accepted by all stages of the detector are regarded as objects. The detection process takes an image as an input and gives at the output the regions that contain the ROI. The false alarm rate of the Haar cascade detector, without a hypothesis verification, is higher but it eliminates most of the non-object regions.

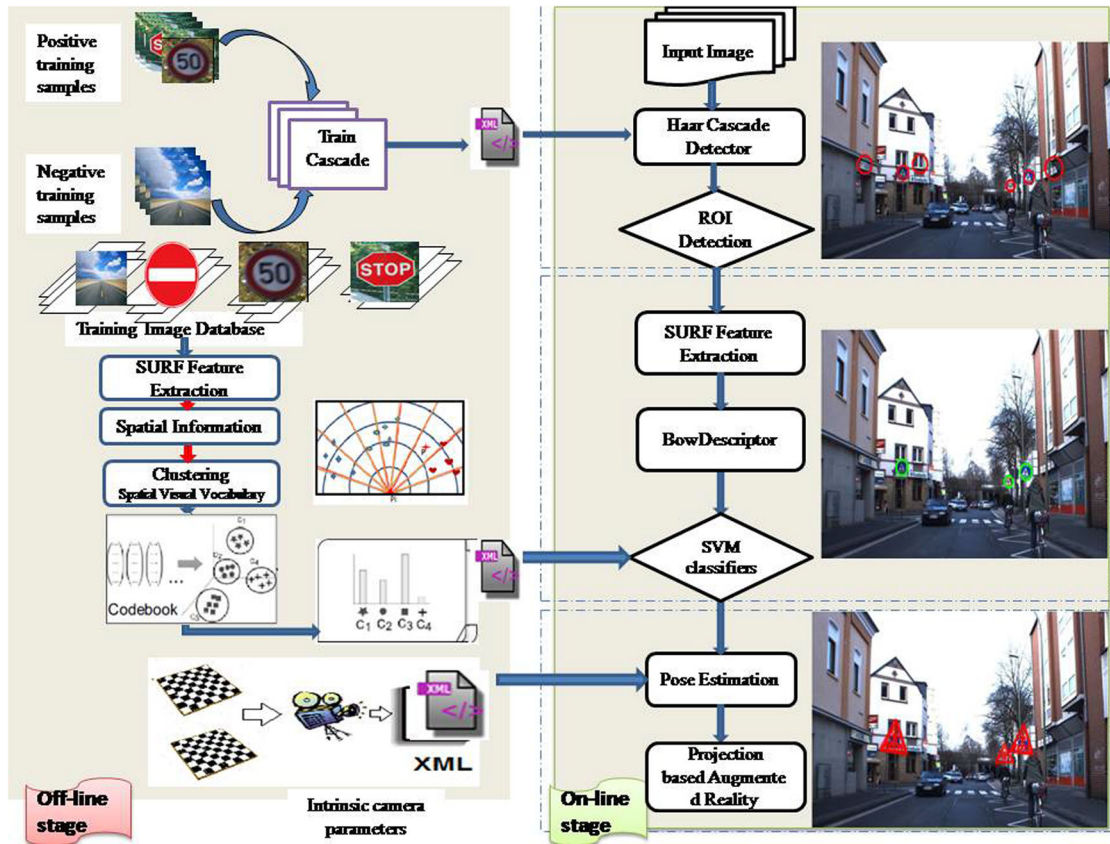


Fig. 1 Overview of the AR-TSR application

The Haar-like features were originally proposed in the framework of object detection in the face detection approach. An AdaBoost cascade using Haar-like features is trained off-line and a boosting algorithm is used to train a classifier with the Haar-like features of positive and negative samples. The AdaBoost Algorithm trains iteratively a strong classifier, which is the sum of several weak classifiers. The object is classified positively only if it is positively classified in each cascade stage. The final classifier works in real time. In fact, from an integral image, in a classifier produced by AdaBoost, voting is done as a summation of weighted classifiers. On average, only a small subset of classifiers vote positively because of cascading.

The real-time capability of the approach is mainly enabled by two properties: most sliding windows are only evaluated by the first stages which contain few classifiers/features (Houben et al. 2013). To reduce the false alarm rate, the detected traffic sign output of the detector stage is processed with a part based on a verification module.

3.2 Verification system based on BoVW

In both instances, the ability to recognize signs and their underlying information is highly desirable. This information can be used to warn the human driver of an oncoming change, or in more intelligent vehicle systems, to actually control the speed and/or steering of the vehicle. It is, therefore, necessary to classify the characteristics of the information given and find a way to represent the information according to these characteristics. BoVW is one of the most successful strategies for object recognition, used to represent an image as a vector of counts using a learned vocabulary. A lot of researchers have worked with natural object recognitions and also used the concept of BoVW for TSR classification and recognition. The BoVW method was inspired by the BOW approach for text categorization (Zagoris et al. 2014). In the BOW, text documents were encoded as a histogram of the number of occurrences of each selected word. One limitation of this approach relies on its inability of encoding spatial distribution of the visual words within an image, which is important for similarity measurement between images (Pedrosa and Traina 2015). To overcome this problem, we proposed a novel approach to integrate the spatial information to BoVW model.

3.2.1 Enhanced BoVW using spatial information

This section details our approach to build an enhanced BoVW using spatial information. In the traditional BoVW model, spatial information between keypoints is ignored during visual word construction when using simple clustering algorithms such as k-mean. However, this modeling approach does not take into consideration the spatial relationships of these words, which is important for similarity measurement between images. To solve this challenging task, recent approaches try to capture information about the relative spatial location of visual words. This paper presents a new approach to integrate the spatial information to BoVW model, with explicit local and global structure models. The key idea is to consider the spatial distribution of visual words in an image. In Khan et al. (2015), a pairwise spatial histogram is defined according to a discretization of the spatial neighborhood into several bins encoding the relative spatial position (distance and angle) of two visual words. Therefore, combining the frequency of occurrence and spatial information of visual words should be a promising direction for improving the image characterization.

Similarly, this paper introduces the outstanding BoVW method for the TSR and makes some developments to integrate spatial information into inverted index to improve accuracy while maintaining short retrieval time, inspired by the study of Nguyen et al. (2014). To address this issue, we introduce a novel way to incorporate both distance and angle information in the BoVW representation. This method exploits spatial orientations and distances of all pairs of similar descriptors in the image. In the BoVWs model, a visual vocabulary $\text{Voc} = v_i, i = \{1, \dots, k\}$, then it is built by clustering these features into a certain number of K visual words. A given descriptor d_k is then mapped to a visual word V_i using euclidean distance 1 as follows:

$$v(d_k) = \operatorname{argmin} \operatorname{Dist}(v, d_k) \quad (1)$$

where $v \in \text{Voc}$, d_k is the k th descriptor in the ROI, $\operatorname{Dist}(v, d_k)$ is the distance between the descriptor and the visual word based on the euclidean distance. For this reason, we consider the weighted sum of ROIs to implicitly represent spatial information which is important for similarity measurement between images.

Since the difference between certain sign types is rather small, a visual dictionary is generated independently for each sign type. Once a dictionary with visual words is generated using the K -means algorithm, a count is made on the frequency of every word, where the vector composed of these frequencies is a codedbook. Despite the effectiveness of ROI classification BoVW model, the accuracy of visual vocabulary directly influences the precision of classification and the size of vocabulary. To represent a ROI in an image, we employ spatial pyramid to enhance voting and indexing criterion of the original inverted indexing technique. Based on the position of each visual word, this method exploits spatial orientations and distances of all pairs of similar descriptors in the image. It is relatively efficient during classification and can well present the spatial information contained in spatial pyramid features.

In the training stage, the SURF features are extracted from all the training samples, using a dense grid. Since we are interested in the sign contents, only the descriptors that do not fall outside the sign contour are taken into account. Our system exploits the SURF features, which have shown a high robustness to varied recording conditions. After the SURF features are extracted for all the training samples, the number of feature points of each image is not entirely consistent, which will bring great difficulties to subsequent operations. Assignment of a visual feature to the vocabulary depends on the similarity metric. We propose a method that incorporates spatial information at feature level. We measure the spatial relationships between visual words using distance and orientation.

3.2.2 Similarity measure of visual words in BoVW

To improve similarity measurement between pairwise, we propose a simple and efficient method to infuse spatial information. These methods are inspired by Zhang and Mayo (2010), using a log-polar quantization of image spatial domain as shown in Fig. 2.

For each visual word, the average position and the standard deviation is computed based on all the occurrences of the visual word in the image. We consider the interaction between visual words by encoding their spatial distances, orientations and alignments. Figure 3 shows an example to better understand our approach. To encode spatial information, we use the distance (3a) and orientation (3b) information between pairs of patches in the image space.

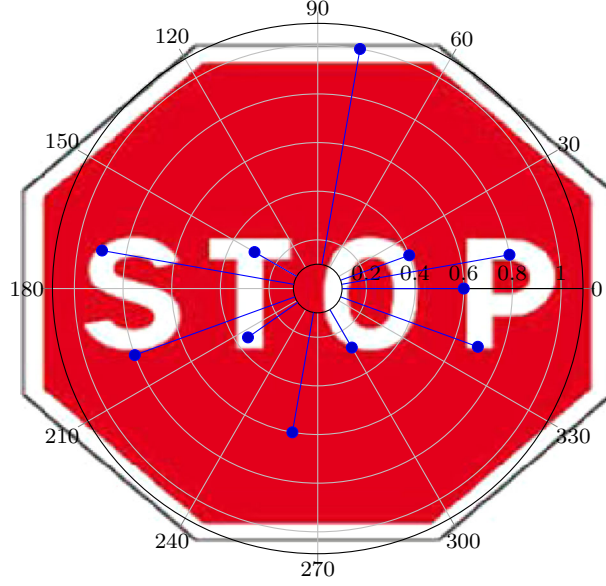


Fig. 2 Log-polar spatial quantization to extract spatial information

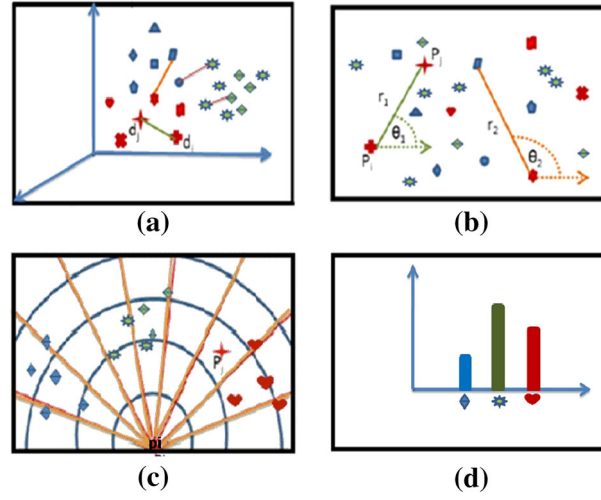


Fig. 3 Spatial histogram of similar pairwise using distance and orientation: **a** spatial distance of similar visual word, **b** spatial orientations of similar pairwise, **c** pairwise similarity distance orientations information of similar patches, and **d** pairwise spatial histograms

More formally, we consider the set S_k of all the pairwise, where at least one patch in the pair belongs to the visual word w_k . A given pair $(P_i, P_j) \in S_k$ is characterized both by a pair of descriptors (d_i, d_j) and a pair of positions in the image space denoted (p_i, p_j) is illustrated in Figure 3. Note that both d_i and p_i are vectors with $d_i \in R^D$ and $p_i \in R^2$.

Then, for each pair of points of the feature, we compute the angle θ formed with the horizontal axis using the following equation:

$$\theta = \begin{cases} \arccos \frac{\overrightarrow{P_i P_j} \cdot \vec{i}}{\|P_i P_j\|} & \text{if } \overrightarrow{P_i P_j} \cdot \vec{j} > 0 \\ \pi - \arccos \frac{\overrightarrow{P_i P_j} \cdot \vec{i}}{\|P_i P_j\|} & \text{otherwise} \end{cases} \quad (2)$$

After clustering the spatial information is implicitly included in the visual vocabulary. A pairwise spatial histogram (3d) of similar patches is then defined considering a discretization of the image space into M bins denoted b_m , $m = \{1, \dots, M\}$ with an angle $\theta \in [0, \pi[$ split into M_θ angle bins and the radius $r \in [0, R]$ split into M_r radial bins so that $M = M_\theta \cdot M_r$.

For those purpose, a novel structural relationship between patches are defined for evaluating super-pixels similarity. Particularly, simple spatial relations between visual words are considered the spatial locations of the words and the spatial relationship between the words was added to describe images in the BoW model. This histogram encodes spatial information (distance and orientation (3d)) of pairwise similar patches, where at least one of the patches belongs to V_k . To have a global representation, we replace each bin of the BoVW frequency histogram with the spatial histogram associated with w_i . By this way, we keep the frequency information intact and add the spatial information. This modularity facilitates simple way to assemble the spatial histograms and to obtain the final representation.

In the next step, a quantization of the descriptor is used by performing a clustering with the k -means algorithm and resulting into a vocabulary of visual words. Therefore, the K -means clustering is applied to each feature descriptor extracted from the local ROI to build the visual vocabulary dictionary, local features are then encoded by calculating their similarities with a dictionary of visual words. The essence of the K -means is to divide disordered data sets into K groups, minimizing the difference of the data within the same group and enlarging it within the different groups as much as possible.

A visual word is expressed by a set of features that corresponds to local image information of the pixels of the image which is identified by its key points (Yeh and Wickens 2001). At this time, spatial information of all features is employed. We use spatial pyramid to partitioning all images into increasingly fine cells depending on pyramid levels defined, and collecting all words found inside each cell. Then, the local features clustered into groups, and each cluster now works as a word of the visual dictionary.

The output of the detection stage is a list of ROI objects that could be probable road signs. One of the biggest challenges on the AR field is determining in real time the virtual image to be displayed, its position and how it should be represented. In the conventional BoVW method, the final representation of the image is a histogram of visual words passed to a linear SVM classifier for classification. If the recognition is complete, a multiclass sign classifier takes the positive ROI and assigns a 3D traffic sign to each one.

Once the ROI is classified, the next step is to estimate the camera motion parameters over time to enhance the image view by associating extra contextual information and enriching the visual content, as well.

3.3 Pose estimation and augmentation

3.3.1 Pose estimation

In a marker-less AR, the problem of finding the camera pose requires significantly more complex and sophisticated algorithms, e.g., disparity mapping, feature detection, extraction, matching and classification. Camera calibration allows combining virtual and real-world objects in a single display. In case of images or videos, the relative position of an element on a screen can be calculated from camera parameters and relative position information of the camera with respect to the element.

To make the information correctly match the real environment according to the variations in the drivers' view, the pose estimation of target ROIs is an essential approach in the AR. Although the problem of camera calibration has been extensively studied, our scope of application imposes some restrictions to the generic problem. More precisely, we are interested in a low-cost resolution for on-board vehicular cameras. To render a virtual object into the real world, a virtual camera has to be placed in the same position and orientation as the real camera. To estimate the camera pose, we perform a camera calibration off-line and store the camera's intrinsic parameters. The intrinsic parameters are those specific to the camera, such as the focal length, principal point, and lens distortion. The extrinsic parameters refer to the 3D position and orientation of the camera.

To estimate the camera's extrinsic parameters for a given frame, some correspondences between 2D points from the image and 3D points from the model are needed. The camera pose can be found from projecting the 3D coordinates of the features into the 2D image coordinate, along with minimizing the difference between their corresponding 2D features. The localization of the object in the real-time image uses the homography matrix to find the corresponding comers of the object in the reference and real-time images. Basically, the system takes four known points from the image of a scene and sets four separate

tracking windows around the points. The first three input parameters, namely *Rotation*, *Scale* and *Distortion*, are extracted directly from the homography. The relationship between camera model and tracking target is shown in Fig. 4.

The key to realize an AR 3D-registration is to obtain a camera projection matrix, which represents the relationship between the 2D points from the image and the 3D points from the model. The geometric relationship between 3D world lines and their projections on the camera image is built to estimate if the relative 6-DOF camera pose consists of rotation parameters and translation parameters (Han and Zhao 2016). From the planar homography, we can easily compute the camera position and rotation, which provide the motion estimates. The used mathematical model is the projection transformation, which is expressed by the following equation:

$$x = \lambda PX = K[R|t]X \quad (3)$$

where λ is the homogeneous scale factors unknown a priori, where P is a 3×4 projection matrix, $x = (x, y)$ are the homogeneous coordinates of the image features, $X = (X, Y, Z)$ are the homogeneous coordinates of the feature points in the world coordinates, $K \in R^{3 \times 3}$ is the matrix with the camera intrinsic parameters, also known as camera matrix, the joint rotation-translation matrix $[R|t]$ is the matrix of extrinsic parameters, $R = [r_x r_y r_z]$ is the 3×3 rotation matrix, and $T = [t]$ is the translation of the camera.

The projection matrix P is the key to creating a realistic augmented scene using the intrinsic parameters of the camera, the dimensions of the video frame, and the distances of the near and far clipping planes from the projection center. In our method, we assume that the intrinsic parameters are known in advance and do not change, and this is reasonable in most cases.

$$P = \underbrace{\begin{pmatrix} 1 & 0 & x_0 \\ 0 & 1 & y_0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{Intrinsicmatrix}} \times \underbrace{\begin{pmatrix} R & t \end{pmatrix}}_{\text{Extrinsicmatrix}} = \underbrace{\begin{pmatrix} 1 & 0 & x_0 \\ 0 & 1 & y_0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{2Dtranslation}} \times \underbrace{\begin{pmatrix} f_x & 0 & 0 \\ 0 & f_y & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{2Dscaling}} \times \underbrace{\begin{pmatrix} 1 & s/f & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{2Dshear}} \times \underbrace{\begin{pmatrix} I & t \end{pmatrix}}_{\text{3Dtranslation}} \times \underbrace{\begin{pmatrix} R & 0 \\ 0 & 1 \end{pmatrix}}_{\text{3Drotation}} \quad (4)$$

Once K is known, the extrinsic parameters for each image are readily computed. From Eq. 3, we have:

$$\begin{aligned} r1 &= \lambda + K^{-1}h_1 \\ r2 &= \lambda + K^{-1}h_2 \\ r3 &= r1 * r2 \\ t &= \lambda + K^{-1}h_3 \end{aligned} \quad \text{where} \quad \left\{ \begin{array}{l} h_1 = [h_{11} \ h_{21} \ h_{31}]^T \\ h_2 = [h_{12} \ h_{22} \ h_{32}]^T \\ h_3 = [h_{13} \ h_{23} \ h_{33}]^T \\ r_1 = [r_{11} \ r_{21} \ r_{31}]^T \\ r_2 = [r_{12} \ r_{22} \ r_{32}]^T \\ r_3 = [r_{13} \ r_{23} \ r_{33}]^T \end{array} \right. \quad \text{where} \quad H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \quad \text{and} \quad \lambda = \frac{1}{\|K^{-1}+h_1\|} \quad (5)$$

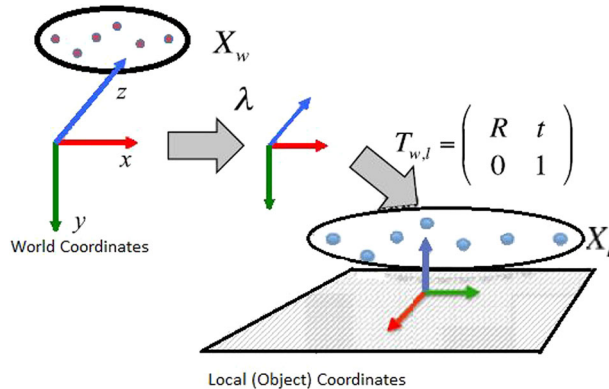


Fig. 4 Coordinate systems involved in camera calibration

To integrate virtual objects into the real-world seamlessly, the AR system must be able to recognize and track its desired environment. However, the marker-less techniques are still challenging and very potential to be explored, so as to develop a seamless AR experience. Inside the final augmented view, virtual objects have been well aligned with real objects in position, orientation, and scale without any visual inconsistency. The vehicle surrounding along the predicted host path and the expected positions of traffic signs have been modeled in 3D, then a projection to the image plane has resulted in a ROI.

3.3.2 Projection-based augmented reality

In this final stage, the projection of virtual objects will be easily accomplished once the pose is known. Having calculated the camera's interior and exterior orientations for a video frame, the 3D can be drawn at the right position, with the proper scale, orientation and perspective in the scene of the real world. With the complete set of camera parameters, virtual objects can be coherently inserted into the video sequence captured by the camera, so that synthetic traffic signs may be added to increase safety.

The projection-based AR corresponds to the use of projection technology to augment and enhance 3D objects and spaces in the real world by projecting images onto their visible surfaces. Once there are enough successful matches, a RANSAC method is applied to calculate the homography matrix between the image of the frame and the image of the object. Then we are able to estimate the 3D pose and draw a virtual 3D object on the top of the real object. The camera calibration allows combining virtual and real-world objects in a single display. In the case of images or videos, the relative position of an element on a screen can be calculated from the camera parameters and relative position information of the camera with respect to the element.

To correctly model the perspective projection of the camera, we must mimic the intrinsic camera parameters in the virtual environment. When we have the camera calibrated in a frame, we can synchronize the real camera with a virtual camera and project the virtual objects onto the real image using OpenGL. Technically, this can be described with a projection matrix that maps 3-D points onto a 2-D plane. After the world has been aligned with the camera using the view transformation, the conversion from an intrinsic matrix to the model view and projection matrices requires a conversion from the world coordinates to the normalized view volume coordinates used by OpenGL. The perspective projection matrix is expressed by the following equation:

$$\begin{bmatrix} x_{clip} \\ y_{clip} \\ z_{clip} \\ w_{clip} \end{bmatrix} = \begin{bmatrix} \frac{2*c_x}{width} & 0 & 1 - \frac{2*x_0}{width} & 0 \\ 0 & \frac{2*c_y}{height} & -1 + \frac{2*y_0}{height} & 0 \\ 0 & 0 & \frac{near+far}{near-far} & -2 * \frac{near*far}{near-far} \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} X_{camera} \\ Y_{camera} \\ Z_{camera} \\ 1 \end{bmatrix} \quad (6)$$

where width, height, far, near represent the positions of the clipping planes.

As it is indicated, most of current marker-less tracking approaches require a 3D model of the environment for matching 2D features to those lying on the model. In addition to the complexity of building a model, such a strategy will result in performance problems when the model is very complex or the environment is dynamic. In contrast, our approach does not need to perform 3D engineering of the environment. Also, we use a simple virtual 3D model, with a known size, to define a reference coordinate system. For the robust tracking of the camera pose, we have developed a new marker-less approach that combines information from both real and virtual worlds. To implement AR-TSR in vehicle, we developed a projection type HUD with 7.5m focal length and 50-inch screen and used a windshield of Genesis DH model as shown in Fig. 5.

This stage is composed of a feature tracker that finds point matches, and a homography-based method can be applied to find the rotation and translation of the camera. Finally the registration matrix is calculated using the above homography, and the virtual objects are rendered on the real scenes using OpenGL. Thus, the appropriate conversions have been made for the proper combination of the obtained results using OpenCV and OpenGL, and the consequent right augmentation of the real-world scenes. Next, we proceed to validate this model via an experiment, conducted under real driving conditions.

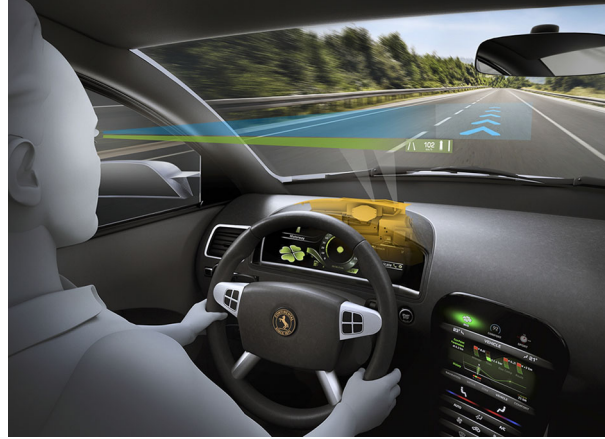


Fig. 5 Projection-based augmented reality

4 Experiment results

Recognizing traffic signs is essential for the automotive-industry efforts in the field of driver assistance and for many other traffic related applications. To demonstrate the performance of our TSR system, some experiments have been performed on realistic videos captured from a moving vehicle. To test the validity of our approach and to ensure a high diversity of road conditions we have tested on a public available database German traffic sign recognition benchmark (GTSRB) and German traffic sign detection benchmark (GTSDb) (Stallkamp et al. 2011), on different days and with different weather conditions.

To evaluate the performance of the proposed algorithm, we implement the proposed AR-TSR method using the hardware environment of Core i7 640LM 2.13 GHz and the software environment of Windows 7, Visual Studio 2010 using OpenGL and OpenCV Library 2.4.8. In the GTSRB dataset, there are 51,839 German traffic signs in 43 classes. These classes of traffic signs have been divided into six subsets: speed limit sign subset, danger sign subset, mandatory sign subset, unique sign subset, derestriction sign subset and other prohibitory sign subset (Yin et al. 2015). The solution of each image in the dataset is 1366×800 . Here, we present quantitative a analysis of the system presented in this work.

4.1 Performance of the proposed method

The first experiment aims to demonstrate the general performance of the proposed system; we separately evaluate the detection and classification modules. To evaluate the suggested detection algorithm and the performance of the classifier, we use image sequences acquired during the daytime in both urban and highway environments under different meteorological conditions.

4.1.1 Detection performance

The database used to train the detectors has been collected from the GTSRB dataset, the Belgian Traffic Signs dataset (BelgiumTS) (Timofte 2017), and our own images. Our training data set consists of 4500 interest traffic signs and 6000 non-traffic signs. The sizes of traffic sign examples are in range from 15×15 to 250×250 pixels (Fig. 6).

Due to the different appearance frequencies of each type of sign and the high intra-class variability, we have trained a detection cascade for each group of similar signs. To get a high-efficiency classifier, we have set the number of cascade stages to 7. The minimum hit rate is set to 0.995 while the maximum false-positive rate is set to 0.5. The samples width and height have been both set to 24×24 . The evaluation is based on the recall and precision values, which are defined as follows:

$$\text{Recall} = \frac{TP}{TP + FN} \times 100 \quad (7)$$

$$\text{Precision} = \frac{TP}{TP + FP} \times 100 \quad (8)$$

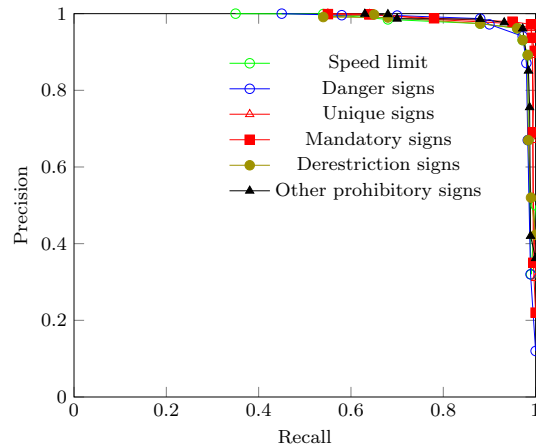


Fig. 6 Precision–recall curve for traffic sign detection

Table 1 Recall and precision results for traffic sign detection

Traffic signs	Number of signs	TP	Recall (%)	Precision (%)
Speed limit	127	126	99.21	98.43
Danger signs	79	75	97.46	98.97
Unique signs	50	49	98	99.04
Mandatory signs	135	134	99.14	99.25
Derestriction signs	120	117	99.02	98.96
Other prohibitory signs	115	115	99.15	99.08

where Tp is true positive, FP is false positive, and FN is false negative.

The achieved detection performances are summarized in Table 1 vs. the number of test images.

The experimental results of Table 1 demonstrate an excellent performance of our system. The results show that the proposed algorithm attains an average precision rate of 98.95 % and an average recall rate of 98.66 %. As previously mentioned, the detection system robustness is demonstrated through its tolerance of changes in lighting and in plane rotations. To evaluate the system robustness, we have tested the accuracy of our algorithm when tracking the ROIs in the captured frames in various lighting and weather conditions, as shown in Fig. 7.

The missing chances of true positives are comparatively less when compared with other systems. The false alarm rate is reduced greatly when the system is tested with a part-based BoVW verification. It has been proved by experiments that our algorithm is not only highly efficient, but also more accurate than previous algorithm during detection. Next, a comparative evaluation of the classifier, based on the proposed discriminative local region representation of traffic signs, is presented.

4.1.2 Classification performance

To put any augmented information the most important thing is to understand the class of the object and identify it properly. The identification of a traffic sign is a challenging task because the traffic sign category number is large and the multi-class recognition problem is not solved. In the classification stage, we determine if a detected image region contains a particular traffic sign or if it has to be rejected as a false positive.

The database used to train the classifiers has been designed using the ROIs obtained from the detection step and the model fitting methods presented in the previous sections. To evaluate the occlusion robustness of the suggested classification method, the content of the detected ROI is identified using the tree classifiers. This classifier is tested on static, low-resolution sign images. Some experiments have also been conducted to measure the performance, the GTSRB has been used, where Table 2 shows the classification rates of the linear SVM.

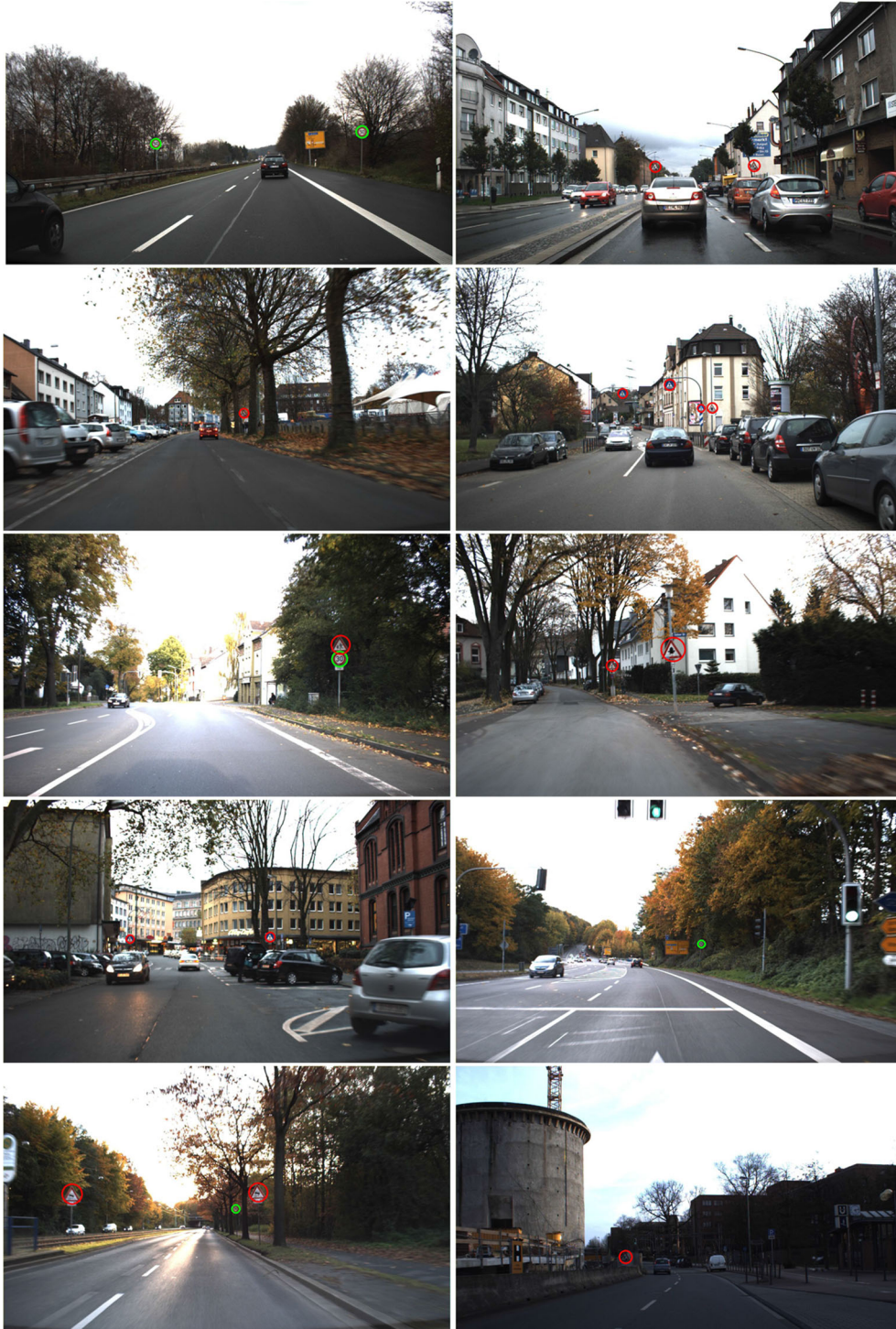


Fig. 7 Detection of traffic signs in adverse conditions

A key idea of our method is to project the 3D object sign using the corresponding sparse dictionary and then to classify the projected vector with the SVM. Furthermore, we evaluate the classification task on the detected signs returned by the previous detection module. As shown in Table 2, the overall classification accuracy is 99.31 %. Note that only 3 (out of 1500) speed limit signs, while only 6 (out of 890) danger signs, are falsely classified. If the recognition is complete, a multiclass sign classifier takes the positive ROI and

Table 2 Accuracy results for traffic sign classification

Traffic signs	Number of signs	TP	FP	Accuracy (%)
Speed limit	1500	1497	3	99.13
Danger signs	890	884	6	98.97
Unique signs	1000	1000	0	99.51
Mandatory signs	1700	1699	1	99.45
Derestriction signs	1350	1347	3	99.32
Other prohibitory signs	1850	1849	2	99.47

Table 3 Performance comparison with other TSR methods

Method	Cireşan et al. (2012)	Stallkamp et al. (2012) (%)	Sermanet and LeCun (2011)	Zaklouta et al. (2011) (%)	Haloi (2015) (%)	Our (%)
Speed limit	99.47	97.63	98.61	95.95	98.82	99.13
Danger signs	99.07%	98.67	98.03	92.08	96.85	98.97
Unique signs	99.22%	100.00	98.63%	98.73	100.00	99.51
Mandatory signs	99.89%	99.72	97.18%	99.27	96.86	99.45
Derestriction signs	99.72%	98.89	94.44%	87.50	97.93	99.32
Other prohibitory signs	99.93%	99.93	99.87%	99.13	98.27	99.47

assigns a 3D traffic sign to each one. Experiments demonstrate that our approach succeeds in adding relative spatial information into the BoVW model by encoding both the global and local relative distribution of visual words over an image.

4.2 Comparisons with other state-of-the-art methods

To verify the discrimination performance and computation efficiency of the proposed feature for TSD, the experiments on the public available data set of traffic signs are implemented. Because the training and testing samples in the GTSRB dataset are split according to a fixed rule, an absolute performance comparison with other reported approaches is possible. We report these results in Table 3, where the results of the winning system from the IJCNN challenge and some reported results in the IJCNN 2011 are provided as references.

The performance results of the machine learning algorithms are all significantly different from each other. We have compared the suggested method with other state-of-the-art algorithms such as the committee of CNNs (Cireşan et al. 2012), human performance (Stallkamp et al. 2012), multi-scale CNNs (Sermanet and LeCun 2011), random forests (Zaklouta et al. 2011; Haloi 2015), wgy@HIT501 (Houben et al. 2013), VISICS (Mathias et al. 2013), LITS1 (Liang et al. 2013), and Viola-Jones. The performance is analyzed in terms of detection and recognition accuracy.

According to the results for the GTSRB data set, shown in Table 3, this work achieves a 99.31% recognition accuracy, which is a comparable performance of 0.24% less than the work by Cireşan et al. (2012), and a performance of 0.17% higher than the work by Stallkamp et al. (2012) and 1.51% than the work by Sermanet and LeCun (2011). The accuracy of recognizing unique signs reach 99.51%, which is comparable with the best achieved one. The danger signs which have triangular shape have given the worst results compared with other traffic sign categories. Compared with other methods, this paper presents an overview of studies related to drivers' perception and cognition when this information is displayed on the windshield HUD, as it can be a solution to reduce the duration and frequency of drivers looking away from the traffic scene, which is very important in safe driving assistance systems.

To verify the processing efficiency and detection accuracy, we have compared the suggested method with other state-of-the-art works including those by Houben et al. (2013), Mathias et al. (2013), Liang et al. (2013) and Viola-Jones. According to the results for the GTSRB data set shown in Table 4, our method achieves high detection accuracy, which reaches 99.61 % area under curve, which is a comparable performance of 0.36% less than the work by wgy@HIT501 (Houben et al. 2013), and a performance of 0.62% higher than the work by VISICS (Mathias et al. 2013), and 2.66 % than the work by LITS1 (Liang et al. 2013).

Considering the training and recognition time (including TSD and TSR), compared with the other works, our hardware platform (Core i7 640LM 2.13 GHz) is not as good as the hardware platform of other methods.

Table 4 Performance comparison with other TSD methods by AREA-UNDER-CURVE (AVERAGE OVERLAP)

Team	Prohibitory (%)	Danger (%)	Mandatory (%)
wgy@HIT501 (Houben et al. 2013)	100 (90.08)	99.91 (86.39)	100 (79.41)
VISICS (Mathias et al. 2013)	100 (88.22)	100 (87.03)	96.98 (89.55)
LITS1 (Liang et al. 2013)	100 (86.91)	98.85 (86.34)	92 (89.49)
Viola-Jones	90.81 (88)	46.26 (84)	44.87 (88)
Our method	99.88 (88.95)	99.27 (89.43)	99.68 (90.93)

Time requirement of the method for 1366×800 images is about 100 ms, which means it is suitable for real-time applications. As it can be seen in Table 3, most of the machine learning algorithms have achieved a correct recognition rate of more than 95 %. Also the CNN-based method attained the highest accuracy rate for all comparison experiments, although the real-time performances are also not satisfying for the ADAS application. These works achieve high detection and classification accuracy, but the time complexity of training and recognition is extremely high.

However, none of the machine learning approaches is able to handle input images of variable sizes and aspect ratios as presented in the dataset. The usual approach is scaling the images to a fixed size, which can cause problems when the aspect ratio is different between the original and target sizes. Furthermore, it discards information in larger images or introduces artifacts if very small images are strongly magnified (Stallkamp et al. 2012).

4.3 Augmented reality tracking

4.3.1 3D overlay

The AR cues may offer a promising means to improve the driver's safety, potentially reducing the response time and increasing the hazard detection likelihood. The stability of tracking is achieved with a temporal smoothing approach that determines the average pose from multiple estimations. Then, the mean pose over the last frames is used. Moreover, we have improved the pose accuracy using a check validation process by computing several poses for each single target localization (Figs. 8, 9, 10, 11).

In this section, the results obtained during real-time tests, performed with a fully equipped vehicle, are presented. We have started the evaluation of the AR tracking by superimposing 3D graphics on target images. To provide driving safety information using the proposed AR-TSR, various sensors and devices have been attached to the experimental test vehicle. The system has been empirically tested under different lighting conditions, during sunny or cloudy days, in the rain and at night.

The experimental results have shown that the proposed method has significantly reduced the computational cost and also stabilized the camera pose estimation process. A virtual object is attached to a real object for the augmentation purpose, and the camera pose is used to superimpose virtual objects onto the real environment. Therefore, the AR-HUD is an important step in the direction of holistic human-machine interaction concepts in vehicles for a more comfortable, more economic, and safer driving experience. The use of the object tracking process keeps the augmentation correct, which proves that pose estimation has been calculated accurately. Moreover, it is able to automatically recognize the scene and identify the patch to be augmented, in spite of changing viewpoint and/or illumination conditions. Using the calculated pose matrix, a virtual 3D object is projected into the real-world scene for augmentation purposes, as shown in Fig. 12.

The experiments have confirmed that the system can accurately superimpose virtual textures or 3D objects to a user-selected planar part of a natural scene in real time, under general motion conditions, without the need of markers or other artificial beacons.

4.3.2 Reprojection error

The calibration of a projector camera system is an essential step toward accurate 3D measurement and environment-aware data projection applications, such as the AR. In this test, we have experimented both methods of pose estimation to reproject 2D points on the image using the computed pose parameters. The aim of this experiment is to evaluate the precision of the pose in reconstructing 2D points from their 3D projections, where $d(\mathbf{x}, \hat{\mathbf{x}})$ denotes the euclidean distance between the image points represented by the vectors $\mathbf{x}$ and $\hat{\mathbf{x}}$.



Fig. 8 Frames illustrating the insertion of virtual 3D object sign



Fig. 9 Insertion of virtual 3D object sign during sunny days

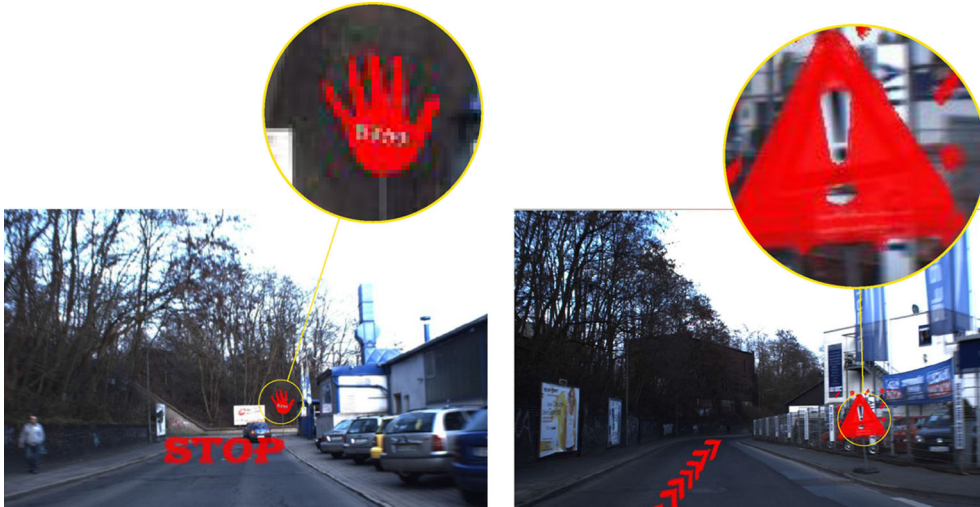


Fig. 10 Insertion of virtual 3D object sign during cloudy days

$$\sum_i d(\mathbf{x}_i, \hat{\mathbf{x}}_i)^2 + d(\mathbf{x}'_i, \hat{\mathbf{x}}'_i)^2 \quad (9)$$

A projector camera system has been set up, and a series of experiments have been conducted to evaluate how the proposed method improves the calibration process of a projector camera system. In addition, we have also evaluated the suggested approach on different calibration patterns with various control point localization algorithms. Figure 13 shows the reprojection error.

As shown in Table 5, the translation error has been very low, 0.28 mm, which is good when dealing with structured lighting applications, as it was the reprojection error. From a technical point of view, displaying detailed 3D information in real time on the HUD has been very challenging. Precisely, projecting 3D arrows that follow the geometry of the road requires a careful calculation of the vehicle position that continuously compensates for the car's movement, in real time. In Fig. 13, we see that our method is more accurate than the PnP method (Wang and Shi 2013). The reprojection error of our method has been 0.34 pixels while it has been 0.96 pixels for the iterative PnP.

The error is computed by reprojecting the 3D coordinates of the object corners onto the 2D space. For each pose, we measure the deviation between the real target corners and the projected corners of the 3D model. We notice that our method is more stable and accurate compared to the PnP algorithm. The latter requires much iteration to converge to the optimal solution.

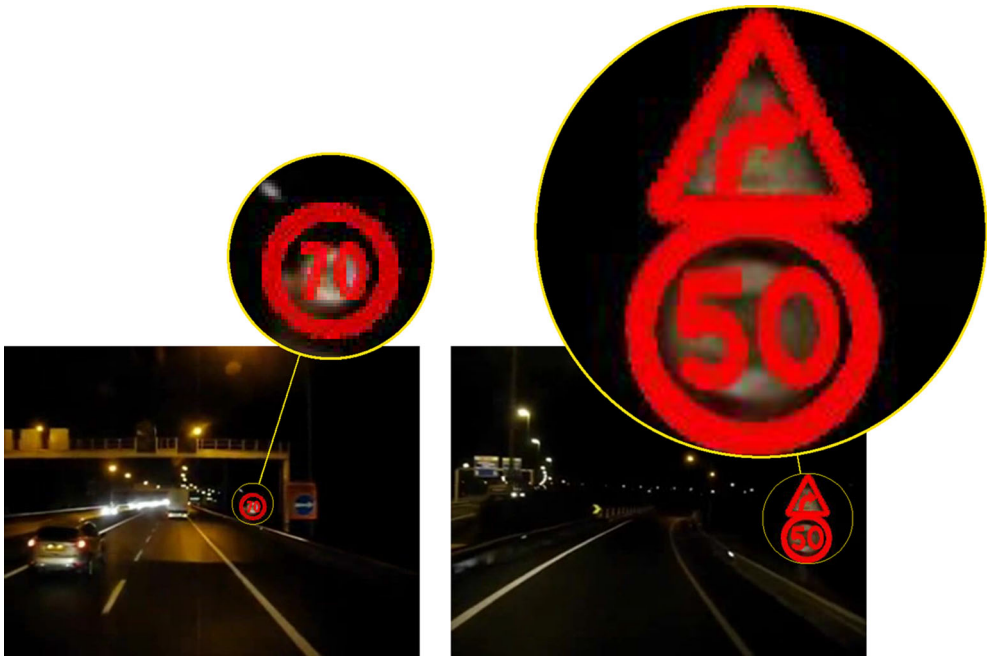


Fig. 11 Insertion of virtual 3D object sign during night time

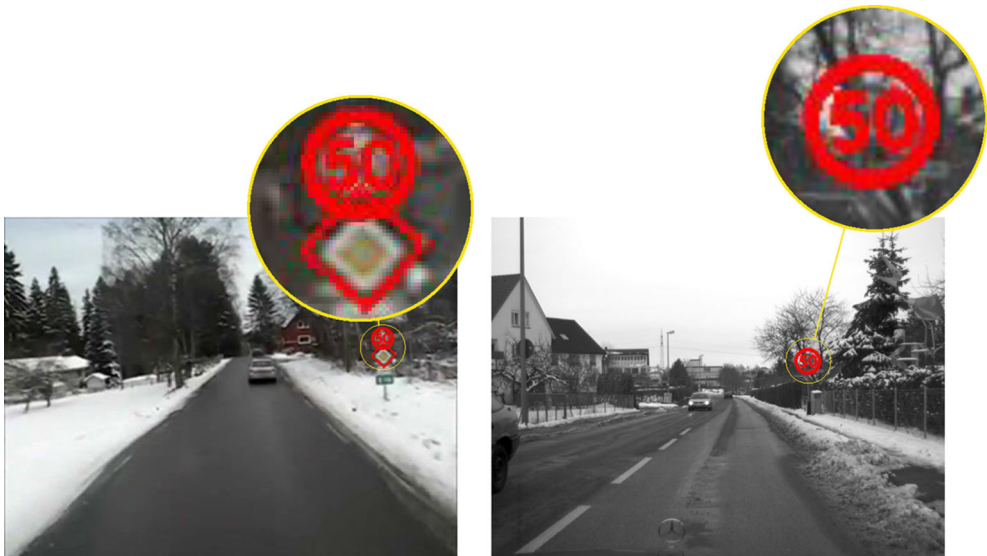


Fig. 12 Insertion of virtual 3D object sign during snowy days

Table 5 Reprojection error

Tested method	PnP method (Wang and Shi 2013)	Our method
Focal length	1.956	1.945
Translation error (mm)	0.74	0.28
Reprojection error	0.96	0.34

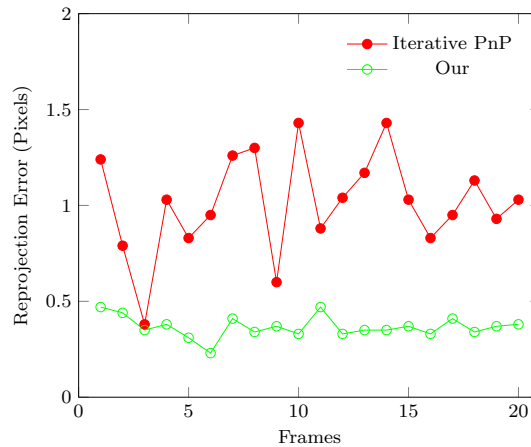


Fig. 13 Reprojection error

5 Conclusions

To improve driving safety and minimize the driving workload, the provided information should be represented in such a way that it is more easily understood and imposes less cognitive load onto the driver. The in-vehicle contextual AR has the potential to provide a novel visual feedback to drivers for an enhanced driving experience. Therefore, the suggested TSR method in this paper presents an overview of studies related to the driver's perception and cognition when this information is displayed on the windshield (HUD), as it can be a solution to reduce the duration and frequency of the driver's looking away from the traffic scene. We have chosen to combine the Haar cascade detector and hypothesis verification using BoVW with the relative spatial information between visual words, which has proved to be a good compromise between the resource efficiency and overall performance.

This study has investigated potential costs and benefits of using dynamic conformal AR cues in improving driving safety. A new AR-HUD approach to create real-time interactive traffic animations is introduced, in terms of rules for placement and visibility, types of traffic signs, and migration of these to an in-vehicle display. The AR-TSR supplements the exterior view of the traffic conditions in front of the vehicle with virtual information for the driver. Here, we have put forward new signs and developed 3D models of these signs, as well as other road infra-structures. First, in the area of vehicular safety, this aims to reduce road accidents through driving assistance systems. Second, we proposed a new computationally efficient method to model global spatial distribution of visual words and improved the standard BoVW representation. It clearly demonstrates the complementarity of the additional relative spatial information provided by our approach to improve accuracy while maintaining short retrieval time. We employ this approach to improve the accuracy of a traffic sign detector to assist the driver in various driving situations, increase the driving comfort and reduce traffic accidents.

Experimental results show that the suggested method could reach comparable performance of the state-of-the-art approaches with less computational complexity and shorter training time. The AR impacts the allocation of visual attention more strongly during the decision-making phase. Indubitably, the speed of our system varies according to different conditions appearing in the video sequences and the complexity of virtual objects, hoping that the display will eventually be used in the windscreens of real cars to help drivers see traffic signs on the road under low-visibility conditions. This is an important next step against driver distraction and sensory overload in the future. The proposed system can perform real-time TSD in sunny or rainy weather or even at night time. AR cues may offer a promising means to improve driver safety, potentially reducing response time and increasing hazard detection likelihood.

Based upon this perspective, the overview made in this paper underlines that the HUD-AR visual display can be a great hope for supporting drivers in terms of enhanced perception and lower workload. In addition, research needs to be conducted on both the amount of information that can minimize a driver's cognitive load and the types of representation that can minimize the drivers visual load. When discussing the AR or other advanced technologies, which will be applied to the next generation of vehicles, it is critical to explore

the deeper cognitive structure information and visualization for these different display types. Our future work is to improve the efficiency of this algorithm, using a GPU acceleration and deep learning to enhance the performance of the proposed system.

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